

Global Stability of Deep Gaussian ReLU Networks

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Abstract

The global Euclidean Lipschitz constant of a square ReLU network with independent Gaussian weights of variance α/n admits an explicit probability bound at every width and depth. The bound follows from a sign-symmetry identity that expresses weighted activation-pattern realizability through the number of orthants met by a formal preactivation subspace, combined with Gaussian radial moments. For each $\alpha < 2$, sufficiently large fixed depth yields global non-expansiveness with exponentially high probability as the width tends to infinity; for $\alpha > 2$, every fixed depth is expansive with exponentially high probability. Consequently, the fixed-depth variance thresholds converge to 2, with a certified lower bound $2 \exp[-(\sqrt{10} + o(1))\sqrt{\log L/L}]$. Biases with independent symmetric coordinates, independent of the weights, satisfy the same stability bound. The symmetry identity also gives exact finite matrix formulas for expected strict activation-region counts and their fixed-width depth asymptotics. For a Gaussian matrix reused at every layer, adaptive Gaussian conditioning gives exponentially concentrated norm propagation over each fixed time horizon, for prescribed inputs and outside a set of small measure under any input distribution independent of the weights.

Keywords: random neural networks, Lipschitz continuity, ReLU activation, Gaussian initialization, activation regions

1 Introduction

Random initialization determines activation scale and sensitivity to input perturbations. For the rectified linear unit, $\phi(t) = \max\{t, 0\}$, weight variance $2/n$ compensates for the loss of half the second moment under rectification (He et al., 2015). Signal-propagation theory relates this normalization to the depth scales over which inputs remain distinguishable (Schoenholz et al., 2017). Bounds on activation-length fluctuations (Hanin and Rolnick, 2018) and expected curve distortion (Hanin et al., 2022) describe prescribed inputs. Global stability requires control of distances between every pair of inputs for a single realization of the weights.

The relevant quantity is the global Euclidean Lipschitz constant. It controls the amplification of arbitrary input perturbations and supplies a sufficient condition for non-expansiveness of an initialized representation. For a piecewise-linear network, this constant is the largest operator norm of a Jacobian on a nonempty activation region. The region and its Jacobian depend on the same weights, so treating the activation mask as independent of the corresponding matrix product is generally invalid. A product of layerwise operator norms avoids that dependence, but does not retain the contraction introduced by the realized ReLU masks.

Consider networks whose input, hidden layers, and output all have dimension n , with ReLU at every layer and independent $N(0, \alpha/n)$ weights. For each $\alpha < 2$, the finite-width bound in Theorem 1 proves global non-expansiveness with exponentially high probability at every sufficiently large fixed depth. For $\alpha > 2$, a prescribed input expands with exponentially high probability at any fixed depth.

The critical variance for activation norms therefore also governs the limiting global stability threshold when width tends to infinity before depth.

The proof uses coordinate sign changes that preserve the joint Gaussian law. For a fixed formal mask, these transformations preserve its Jacobian norm and move its realizability constraint through every coordinate orthant. Averaging gives an exact identity for the expectation of any nonnegative invariant functional multiplied by the realizability indicator. A hyperplane-arrangement bound (Cover, 1965) reduces the counting factor from 2^{nL} to a quantity with logarithm of order $n \log L$. A sphere net and exact moments of a Bernoulli mixture of Gaussian squares complete the estimate.

The same identity yields exact formulas for the mean number of strict activation regions. A triangular transfer matrix of size depending only on the width gives the count at every depth. At fixed width, the zero-bias mean and the probability that the network map is not identically zero decay at rate $(1 - 2^{-n})^L$. With independent nondegenerate centered Gaussian biases, the strict-region mean instead increases to a finite limit. Throughout, distinct activation patterns are counted separately, including patterns with identical final output maps.

For the tied-weight iteration $x_{t+1} = \phi(Wx_t)$, adaptive orthogonal queries isolate independent Gaussian innovations. Concentration over their finite-dimensional coefficient space gives an exponential deviation bound for the norm profile over a fixed horizon, with constants uniform over deterministic unit starts. Integration over an input distribution independent of the matrix gives a bound on the measure of exceptional inputs. Simultaneous control over all starting vectors requires a global estimate beyond this finite-horizon argument.

1.1 Related Work

Global sensitivity of random networks. Geuchen et al. (2025) bound the Lipschitz constant of scalar-output Gaussian ReLU networks. Their deep upper bound assumes hidden width $N > d + 2$, where d is the input dimension, and their lower bound requires width of order at least dL^2 . Dirksen et al. (2025) obtain sharper bounds for ℓ^p input metrics. Their Theorem 4.1 also gives an absolute-constant Euclidean Lipschitz bound for vector-valued intermediate maps at weight variance $2/N$, under width conditions of order dL^3 up to logarithmic factors and with $N \geq Cd$ for a sufficiently large constant C . These width assumptions exclude the square regime $d = N = n$ of Theorem 2.

Sign symmetry gives independent Bernoulli-mask representations for Jacobian quantities at prescribed inputs (Bartlett et al., 2021, Lemma 2.1); Dirksen et al. (2025, Proposition 2.11) give a more general distributional identity. Proposition 5 weights a Jacobian functional by the existence of a strict activation region. Its orthant count provides the global comparison used in Theorem 1.

Activation geometry and collapse. Hanin and Rolnick (2019a) study the complexity of linear regions along one-dimensional subspaces and the volume of region boundaries. Hanin and Rolnick (2019b) bound expected activation-pattern counts in bounded input sets under assumptions on gradients and bias densities. Wang (2022) compares expected maximal linear-region counts for several architectures, while Kogan et al. (2025) studies output breakpoints for scalar-input, scalar-output networks as hidden widths grow. Theorem 15 counts strict activation patterns on all of $\mathbb{R}^n$ at finite width and depth. The transfer formula combines hyperplane-arrangement identities (Stanley, 2007; Zaslavsky, 1975) with an almost-sure rank recurrence for the formal network stack.

Whole-network death, fixed-input survival, and normalized survival probabilities are analyzed by Lu et al. (2020) and Rister and Rubin (2020). The exact transfer formula in Theorem 15 yields a

finite positive normalized survival limit on the full square input space, as well as an explicit leading coefficient for the mean strict-region count.

Repeated Gaussian matrices. Gaussian conditioning controls adaptive matrix computations in approximate message passing (Bayati and Montanari, 2011). The Tensor Programs limit theorem (Yang, 2019, Theorem 5.4) gives Gaussian limits and limiting norm recursions for recurrent architectures with shared weights. Theorem 20 supplies a finite-width deviation bound for the specified ReLU iteration: its failure probability is Ce^{-cN} for a fixed horizon, with constants uniform over deterministic unit starts.

1.2 Scope and Organization

The global bound certifies sensitivity control at initialization for variances below $2/n$. Its sufficient depth can be large near $2/n$, and activation norms shrink in this regime. Representation quality, trainability, and stability after weight updates require separate analysis. The fixed-depth threshold bound is one-sided; a matching asymptotic formula remains open.

Section 2 states the global bounds. Sections 3 and 4 prove them through the weighted sign-symmetry identity and radial moments. Section 5 derives the exact region-count formulas, and Section 6 treats tied weights. Section 7 reports reproducible numerical evaluations. Appendix A establishes the exact scalar moment exponent and the depth-one threshold.

2 Model and Main Stability Bounds

For integers $n, L \geq 1$ and $\alpha > 0$, let $W_1, \dots, W_L$ be independent $n \times n$ matrices whose entries are independent $N(0, \alpha/n)$ random variables. Set

$$x_0(s) = s, \quad h_\ell(s) = W_\ell x_{\ell-1}(s), \quad x_\ell(s) = \phi(h_\ell(s)), \quad F_{n,L}^{(\alpha)}(s) = x_L(s), \quad (1)$$

where ϕ acts coordinatewise. Unless biases are explicitly specified, the network is bias-free. Write

$$K_{n,L}(\alpha) = \sup_{s \neq t} \frac{\|F_{n,L}^{(\alpha)}(s) - F_{n,L}^{(\alpha)}(t)\|}{\|s - t\|}. \quad (2)$$

All vector norms are Euclidean, and $\|A\|_{2 \rightarrow 2}$ denotes the induced operator norm. Positive homogeneity gives $F_{n,L}^{(\alpha)}(cs) = cF_{n,L}^{(\alpha)}(s)$ for $c \geq 0$ and $F_{n,L}^{(\alpha)}(0) = 0$. ReLU is 1-Lipschitz, so $K_{n,L}(\alpha) \leq \prod_\ell \|W_\ell\|_{2 \rightarrow 2} < \infty$. Continuity allows the supremum in (2) to be restricted to rational pairs, which proves measurability.

Let $H(u) = -u \log u - (1-u) \log(1-u)$, with $0 \log 0 = 0$, and, for $\theta > 0$, define

$$z_\theta = \frac{2}{1 + \sqrt{1 + 8\theta}}, \quad (3)$$

$$g_\alpha(\theta) = \theta \log \frac{\alpha}{ez_\theta^2(1+z_\theta)} + \log \frac{1+z_\theta}{2z_\theta}, \quad (4)$$

$$B_L(\alpha, \theta) = LH(1/L) + \log 5 + \theta \log 4 + Lg_\alpha(\theta), \quad L \geq 2. \quad (5)$$

All logarithms are natural.

Theorem 1 (Finite-width stability bound) *For all $n \geq 1$, $L \geq 2$, $\alpha > 0$, and $\theta > 0$,*

$$\mathbb{P}(K_{n,L}(\alpha) > 1) \leq \mathbb{E}K_{n,L}(\alpha)^{2\theta n} \leq 2\exp\{nB_L(\alpha, \theta)\}. \quad (6)$$

For $L = 1$, the same bound holds with $LH(1/L)$ replaced by $\log 2$. If $c_L(\alpha) := \sup_{\theta > 0} [-B_L(\alpha, \theta)] > 0$, then $\mathbb{P}(K_{n,L}(\alpha) > 1) \leq 2e^{-c_L(\alpha)n}$ for every width.

For fixed L , define

$$\alpha_L = \sup\{\alpha > 0 : \mathbb{P}(K_{n,L}(\alpha) \leq 1) \longrightarrow 1 \text{ as } n \rightarrow \infty\}. \quad (7)$$

The set is nonempty: a sufficiently small α makes the product of the L Gaussian operator norms less than one with probability tending to one. Coupling $W_\ell^{(\alpha)} = \sqrt{\alpha}W_\ell^{(1)}$ gives $K_{n,L}(\alpha) = \alpha^{L/2}K_{n,L}(1)$, so this set is an interval. Its endpoint need not belong to it.

Theorem 2 (Depth limit of the global threshold) *The network in (1) satisfies the following statements.*

(i) *For each $0 < \alpha < 2$, there is $L_0(\alpha)$ such that every fixed $L \geq L_0(\alpha)$ admits $c_{\alpha,L} > 0$ with $\mathbb{P}(K_{n,L}(\alpha) > 1) \leq 2e^{-c_{\alpha,L}n}$ for all n .*

(ii) *For $\alpha > 2$ and all $n, L \geq 1$,*

$$\mathbb{P}(K_{n,L}(\alpha) \leq 1) \leq \frac{5\alpha^2 L}{n(\alpha - 2)^2}. \quad (8)$$

In fact, the exponential refinement (11) below holds.

(iii) *The fixed-depth thresholds satisfy $\alpha_L \leq 2$ and $\lim_{L \rightarrow \infty} \alpha_L = 2$.*

(iv) *As $L \rightarrow \infty$,*

$$\alpha_L \geq 2 \exp \left[-(\sqrt{10} + o(1)) \sqrt{\frac{\log L}{L}} \right]. \quad (9)$$

The limit in (7) fixes depth before sending width to infinity. The finite estimate (6) can also be evaluated in a joint regime $L = L(n)$; the threshold theorem asserts only the stated order of limits. At depth one, $\alpha_1 = 1/4$, as proved in Appendix A.

For $0 < c < 1/2$, define

$$I(c) = \sup_{s>0} \left\{ -sc - \log \frac{1 + (1+2s)^{-1/2}}{2} \right\} > 0. \quad (10)$$

The sharper expansive bound is

$$\mathbb{P}(K_{n,L}(\alpha) \leq 1) \leq Le^{-nI(1/\alpha)}, \quad \alpha > 2. \quad (11)$$

Proposition 12 proves this estimate from the same scalar distribution used in Theorem 1. The symmetric-bias extension in Section 4 preserves the stability bound without any bound on the size or moments of the biases.

3 Realizable Activation Patterns and the Global Lipschitz Constant

For fixed n, L, α , abbreviate $F = F_{n,L}^{(\alpha)}$ and $K = K_{n,L}(\alpha)$.

3.1 Formal Masks and Strict Activation Regions

A *formal mask sequence* is a tuple $D = (D_1, \dots, D_L)$ of diagonal 0/1 matrices of size n . For a fixed sequence, define

$$H_1^D = W_1, \quad H_\ell^D = W_\ell D_{\ell-1} H_{\ell-1}^D \quad (2 \leq \ell \leq L), \quad J_D = D_L H_L^D. \quad (12)$$

Set $E_\ell^D = 2D_\ell - I_n$. The corresponding strict activation region is

$$R_D = \{s \in \mathbb{R}^n : E_\ell^D H_\ell^D s > 0 \text{ coordinatewise for every } 1 \leq \ell \leq L\}. \quad (13)$$

It is an open convex cone, possibly empty. Induction through the layers shows that, on R_D , the network has preactivations $H_\ell^D s$, activations $D_\ell H_\ell^D s$, and output $J_D s$. Conversely, every input with no zero preactivation coordinate lies in exactly one such region.

Stack the formal preactivations into the matrix

$$A_D = \begin{pmatrix} H_1^D \\ \vdots \\ H_L^D \end{pmatrix} \in \mathbb{R}^{nL \times n}, \quad O_D = \{(y_1, \dots, y_L) : E_\ell^D y_\ell > 0 \text{ for every } \ell\}. \quad (14)$$

Thus

$$R_D \neq \emptyset \iff \text{im}(A_D) \cap O_D \neq \emptyset. \quad (15)$$

These events are measurable. If $\tau_D \in \{\pm 1\}^{nL}$ is the sign vector of O_D , then

$$f_D(A) = \max_{\|s\|=1} \min_i (\tau_D)_i (As)_i$$

is continuous, and the event in (15) is $\{f_D(A_D) > 0\}$.

Lemma 3 (Piecewise affine Lipschitz constant) *Suppose a continuous map $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is affine on each cell of a finite polyhedral subdivision of $\mathbb{R}^n$. If M_C is its linear part on a full-dimensional cell C , then*

$$\text{Lip}_2(f) = \max_C \|M_C\|_{2 \rightarrow 2}.$$

Proof. Let M denote the displayed maximum. A line segment from s to t admits a finite subdivision whose subsegments lie in closures of cells. On each subsegment, continuity identifies f with an affine formula from an adjacent full-dimensional cell. Its change in norm is at most M times the length of that subsegment. Summing, and using additivity of length along the segment, gives $\|f(t) - f(s)\| \leq M\|t - s\|$. Conversely, for an interior point s of a cell C and a unit vector v attaining $\|M_C v\| = \|M_C\|_{2 \rightarrow 2}$, both s and $s + hv$ remain in C for sufficiently small $|h|$. The corresponding difference quotient is $\|M_C\|_{2 \rightarrow 2}$. $\blacksquare$

Lemma 4 (Exact reduction to strict regions) *Almost surely,*

$$K = \max(\{0\} \cup \{\|J_D\|_{2 \rightarrow 2} : R_D \neq \emptyset\}). \quad (16)$$

Proof. Successively subdividing by the preactivation signs produces a finite polyhedral subdivision on which F is linear. To account for preactivations that vanish identically, fix D , put $B_0^D = I_n$ and $B_\ell^D = D_\ell H_\ell^D$, and condition on $W_1, \dots, W_{\ell-1}$. If $B_{\ell-1}^D \neq 0$, each row of $H_\ell^D = W_\ell B_{\ell-1}^D$ is a nonzero linear functional almost surely: a Gaussian row belongs to the proper linear subspace $\{w : w^T B_{\ell-1}^D = 0\}$ with probability zero. There are finitely many masks, layers, and rows, so this assertion holds simultaneously. If $B_{\ell-1}^D = 0$, all subsequent formal products, including J_D , vanish. Consequently, every full-dimensional cell with a nonzero linear part contains an open subset of a strict region with that same linear part; the remaining cells have zero linear part. Apply Lemma 3. ■

3.2 A Weighted Identity for Realizability

For diagonal sign matrices $\Sigma_1, \dots, \Sigma_L$, put $\Sigma_0 = I_n$ and define

$$W_\ell^\Sigma = \Sigma_\ell W_\ell \Sigma_{\ell-1}, \quad S_\Sigma = \text{diag}(\Sigma_1, \dots, \Sigma_L). \quad (17)$$

The 2^{nL} choices form a finite group $\mathcal{G}$ preserving the joint distribution of the weights. Since diagonal masks commute with diagonal sign matrices, induction gives

$$H_\ell^D(W^\Sigma) = \Sigma_\ell H_\ell^D(W), \quad A_D(W^\Sigma) = S_\Sigma A_D(W), \quad J_D(W^\Sigma) = \Sigma_L J_D(W). \quad (18)$$

In particular, $\|J_D\|_{2 \rightarrow 2}$ and $\|J_D v\|$ for fixed v are invariant under this action. The action is applied to the formal products; it does not assert an equivariance of the nonlinear ReLU network under hidden coordinate sign changes.

For a linear or affine subspace $U \subseteq \mathbb{R}^M$, let $r(U)$ be the number of open coordinate orthants meeting U .

Proposition 5 (Weighted sign-symmetry identity) *Fix a formal mask sequence D . For every non-negative measurable functional $G(W_1, \dots, W_L)$ invariant under (17),*

$$\mathbb{E}[G \mathbf{1}_{\{R_D \neq \emptyset\}}] = \mathbb{E}\left[G \frac{r(\text{im} A_D)}{2^{nL}}\right]. \quad (19)$$

Proof. Distributional invariance and (18) imply, for every $\Sigma \in \mathcal{G}$,

$$\mathbb{E}[G \mathbf{1}_{\{R_D \neq \emptyset\}}] = \mathbb{E}[G \mathbf{1}_{\{S_\Sigma \text{im}(A_D) \cap O_D \neq \emptyset\}}].$$

For any fixed U , the sum of the indicators on the right over Σ equals $r(U)$: the orthants $S_\Sigma O_D$ run through every open coordinate orthant exactly once, and $S_\Sigma^{-1} = S_\Sigma$. Averaging proves the identity, also for infinite expectations by nonnegativity. ■

Lemma 6 (Central arrangement bound) *If $U \subseteq \mathbb{R}^M$ is a linear subspace of dimension d , then*

$$r(U) \leq 2 \sum_{j=0}^{d-1} \binom{M-1}{j}.$$

An empty sum is zero, and binomial coefficients outside their range are zero. In particular, define

$$C_{n,L} = 2 \sum_{j=0}^{n-1} \binom{nL-1}{j}. \quad (20)$$

Then $r(\text{im}A_D) \leq C_{n,L}$, $C_{n,1} = 2^n$, and

$$C_{n,L} \leq 2 \exp\{nLH(1/L)\} \quad (L \geq 2), \quad (21)$$

where $H(u) = -u \log u - (1-u) \log(1-u)$ with $0 \log 0 = 0$.

Proof. If a coordinate functional vanishes on U , then $r(U) = 0$. Otherwise the coordinate hyperplanes restrict to a central arrangement on U , and its regions are exactly the nonempty intersections with open orthants. The maximal number of regions satisfies $R(M, d) \leq R(M-1, d) + R(M-1, d-1)$, with $R(M, 1) \leq 2$ and $R(1, d) \leq 2$. Induction using Pascal's identity gives the bound (Cover, 1965). The case $d = 0$ is immediate.

For integers $0 \leq k \leq N/2$, the binomial expansion at $q = k/N$ gives

$$\sum_{j=0}^k \binom{N}{j} \leq \exp\{NH(k/N)\}.$$

For $k > 0$, retain terms $j \leq k$ in $(q + (1-q))^N$ and use $q^j(1-q)^{N-j} \geq q^k(1-q)^{N-k}$; $k = 0$ is immediate. Take $N = nL - 1$, $k = n - 1$, and use $(n-1)/(nL-1) \leq 1/L \leq 1/2$ and monotonicity of H on $[0, 1/2]$. ■

Corollary 7 (Comparison with independent masks) *Let D be uniform over the formal mask sequences and independent of the weights. For every $t \geq 0$ and $r > 0$,*

$$\mathbb{P}(K > t) \leq C_{n,L} \mathbb{P}(\|J_D\|_{2 \rightarrow 2} > t), \quad (22)$$

$$\mathbb{E}K^r \leq C_{n,L} \mathbb{E}\|J_D\|_{2 \rightarrow 2}^r. \quad (23)$$

Proof. Lemma 4 bounds the first event by the union over D of $\{R_D \neq \emptyset, \|J_D\|_{2 \rightarrow 2} > t\}$ and gives $K^r \leq \sum_D \mathbf{1}_{\{R_D \neq \emptyset\}} \|J_D\|_{2 \rightarrow 2}^r$. Apply Proposition 5 and Lemma 6 to each term. Summing over all 2^{nL} masks gives the uniform-mask expectation. ■

4 Radial Moments and the Finite-width Bound

4.1 Reduction to One Scalar Random Variable

A volume comparison supplies a $1/2$ -net $\mathcal{N}$ of the unit sphere with $|\mathcal{N}| \leq 5^n$: take a maximal $1/2$ -separated subset and compare its disjoint radius- $1/4$ balls with the radius- $5/4$ ball. For every linear map T ,

$$\|T\|_{2 \rightarrow 2} \leq 2 \max_{v \in \mathcal{N}} \|Tv\|, \quad \|T\|_{2 \rightarrow 2}^{2p} \leq 4^p \sum_{v \in \mathcal{N}} \|Tv\|^{2p} \quad (p > 0). \quad (24)$$

The first inequality follows by approximating a unit vector within $1/2$ and absorbing $\|T\|_{2 \rightarrow 2}/2$; the second follows by taking powers and bounding a maximum by a sum. These are standard sphere-net estimates (Vershynin, 2018).

Lemma 8 (Exact radial moments) *Fix a formal mask D , let $m_\ell = \text{tr } D_\ell$, and let v be a deterministic unit vector. For every $p > 0$,*

$$\mathbb{E}\|J_D v\|^{2p} = \prod_{\ell=1}^L \mathbb{E} \left[\left(\frac{\alpha}{n} \chi_{m_\ell}^2 \right)^p \right], \quad (25)$$

with $\chi_0^2 = 0$.

Proof. Let R_ℓ retain the coordinates selected by D_ℓ , so that $D_\ell = R_\ell^T R_\ell$. The active blocks $G_1 = R_1 W_1$ and $G_\ell = R_\ell W_\ell R_{\ell-1}^T$ for $\ell \geq 2$ are independent Gaussian matrices of sizes $m_\ell \times m_{\ell-1}$, where $m_0 = n$, and $\|J_D v\| = \|G_L \cdots G_1 v\|$. For an independent Gaussian block G with m rows and a fixed vector y , $\|Gy\|^2$ has distribution $(\alpha/n)\|y\|^2 \chi_m^2$. Condition on the preceding blocks and iterate this identity. If a mask has rank zero, both sides vanish. $\blacksquare$

Let $B_1, \dots, B_n$ be independent Bernoulli(1/2) variables and $Z_1, \dots, Z_n$ independent standard Gaussian variables, independent of the B_i , and define

$$X_n^{(\alpha)} = \frac{\alpha}{n} \sum_{i=1}^n B_i Z_i^2. \quad (26)$$

Conditioning on $\sum_i B_i$ gives

$$\mathbb{E}[(X_n^{(\alpha)})^p] = 2^{-n} \sum_{m=0}^n \binom{n}{m} \mathbb{E}\left[\left(\frac{\alpha}{n} \chi_m^2\right)^p\right]. \quad (27)$$

Proposition 9 (Master moment inequality) *For all $n, L \geq 1$, $\alpha > 0$, and $p > 0$,*

$$\mathbb{P}(K_{n,L}(\alpha) > 1) \leq \mathbb{E}K_{n,L}(\alpha)^{2p} \leq C_{n,L} 5^n 4^p \left(\mathbb{E}[(X_n^{(\alpha)})^p]\right)^L. \quad (28)$$

Proof. Apply (24) inside (23). Conditional on the uniform masks, Lemma 8 applies. Their active counts are independent Binomial($n, 1/2$) variables, so (27) shows that, for each fixed unit v ,

$$\mathbb{E}\|J_D v\|^{2p} = \left(\mathbb{E}[(X_n^{(\alpha)})^p]\right)^L.$$

Sum over the net and use Markov's inequality. $\blacksquare$

4.2 The Scalar High-moment Bound

For $Y = BZ^2$ with independent $B \sim \text{Bernoulli}(1/2)$ and $Z \sim N(0, 1)$, direct Gaussian integration gives

$$\lambda(t) := \log \mathbb{E}e^{tY} = \log \frac{1 + (1 - 2t)^{-1/2}}{2} \quad (t < 1/2), \quad \mathbb{E}Y = \frac{1}{2}, \quad \text{Var}(Y) = \frac{5}{4}. \quad (29)$$

Lemma 10 (Optimized finite-width scalar bound) *For $\theta > 0$, put*

$$z_\theta = \frac{2}{1 + \sqrt{1 + 8\theta}}, \quad g_\alpha(\theta) = \theta \log \frac{\alpha}{ez_\theta^2(1 + z_\theta)} + \log \frac{1 + z_\theta}{2z_\theta}. \quad (30)$$

Then, for every $n \geq 1$,

$$\mathbb{E}[(X_n^{(\alpha)})^{\theta n}] \leq \exp\{ng_\alpha(\theta)\}. \quad (31)$$

Moreover, g_α extends continuously to $g_\alpha(0) = 0$, and

$$g'_\alpha(\theta) = \log \frac{\alpha}{z_\theta^2(1 + z_\theta)}, \quad (32)$$

$$g_\alpha(\theta) = \theta \log(\alpha/2) + \frac{5}{2}\theta^2 + O(\theta^3) \quad (\theta \downarrow 0). \quad (33)$$

The remainder in (33) is independent of α .

Proof. For $x \geq 0$, $p > 0$, and $t > 0$, maximization of $x^p e^{-tx}$ gives $x^p \leq (p/(et))^p e^{tx}$. Apply this with $x = \sum_i B_i Z_i^2$ and $p = \theta n$. For $0 < t < 1/2$,

$$\frac{1}{n} \log \mathbb{E}[(X_n^{(\alpha)})^{\theta n}] \leq \theta \log \frac{\alpha \theta}{et} + \lambda(t).$$

The right side is strictly convex in t , since its second derivative is $\theta/t^2 + \lambda''(t) > 0$, and tends to infinity at both endpoints. Writing $z = \sqrt{1-2t}$ gives

$$\lambda'(t) = \frac{1}{z^2(1+z)}, \quad t\lambda'(t) = \frac{1-z}{2z^2}.$$

The stationary equation $t\lambda'(t) = \theta$ has the unique solution $z = z_\theta$, and substitution yields (30). Differentiating the minimized expression gives $g'_\alpha(\theta) = \log(\alpha\theta/t)$, which is (32). Since $z_\theta = 1 - 2\theta + O(\theta^2)$, this derivative equals $\log(\alpha/2) + 5\theta + O(\theta^2)$. Integrating proves (33). All dependence on α is in the linear term $\theta \log \alpha$. $\blacksquare$

4.3 Finite-width and Threshold Conclusions

Proof of Theorem 1. Take $p = \theta n$ in Proposition 9, and apply Lemma 10. For $L \geq 2$, (21) gives

$$\mathbb{E}K_{n,L}(\alpha)^{2\theta n} \leq 2 \exp\{n[LH(1/L) + \log 5 + \theta \log 4 + Lg_\alpha(\theta)]\}.$$

This is the asserted budget $B_L(\alpha, \theta)$. For $L = 1$, use $C_{n,1} = 2^n \leq 2e^{n \log 2}$ instead. $\blacksquare$

For $L \geq 2$, the budget can be optimized without numerical differentiation. If $\alpha 4^{1/L} < 2$, the equation

$$z^2(1+z) = \alpha 4^{1/L}, \quad \theta = \frac{1-z}{2z^2} \tag{34}$$

has a unique solution $z \in (0, 1)$ and gives its unique minimizer, since $\partial_\theta B_L = \log 4 + Lg'_\alpha(\theta)$ is strictly increasing, negative at zero, and tends to infinity. At that minimizer,

$$\inf_{\theta > 0} B_L(\alpha, \theta) = LH(1/L) + \log 5 + L \left[-\frac{1-z}{2z^2} + \log \frac{1+z}{2z} \right]. \tag{35}$$

If $\alpha 4^{1/L} \geq 2$, the infimum is the positive boundary value $LH(1/L) + \log 5$. In particular, when $c_L(\alpha) := -\inf_{\theta > 0} B_L(\alpha, \theta) > 0$, Theorem 1 gives $\mathbb{P}(K_{n,L}(\alpha) > 1) \leq 2e^{-nc_L(\alpha)}$.

Proposition 11 (Expansion at a fixed input) For every $n, L \geq 1$ and $\alpha > 2$,

$$\mathbb{P}(K_{n,L}(\alpha) \leq 1) \leq \frac{5\alpha^2 L}{n(\alpha - 2)^2}. \tag{36}$$

Proof. Fix a deterministic nonzero input s . Conditional on the preceding layers and on $x_{\ell-1}(s) \neq 0$,

$$\frac{\|x_\ell(s)\|^2}{\|x_{\ell-1}(s)\|^2} \stackrel{d}{=} X_n^{(\alpha)}. \tag{37}$$

This follows because a centered Gaussian's sign is independent of its magnitude. If each ratio is greater than one, the trajectory remains nonzero and $\|F(s)\| > \|s\|$. Since $F(0) = 0$, this implies

$K > 1$. Otherwise there is a first layer with a ratio at most one; this includes the first layer at which the trajectory becomes zero. A union bound over this first failure, using (37), gives

$$\mathbb{P}(K \leq 1) \leq L\mathbb{P}\left(\sum_{i=1}^n B_i Z_i^2 \leq \frac{n}{\alpha}\right).$$

The sum has mean $n/2$ and variance $5n/4$. Chebyshev's inequality therefore gives (36). $\blacksquare$

Proposition 12 (Exponential expansion) *The network in (1) satisfies*

$$\mathbb{P}(K_{n,L}(\alpha) \leq 1) \leq Le^{-nI(1/\alpha)}, \quad (38)$$

for all $n, L \geq 1$ and $\alpha > 2$, with I defined in (10).

Proof. The proof of Proposition 11 gives

$$\mathbb{P}(K_{n,L}(\alpha) \leq 1) \leq L\mathbb{P}(S_n \leq n/\alpha), \quad S_n = \sum_{i=1}^n B_i Z_i^2.$$

For $s > 0$,

$$\mathbb{E}e^{-sS_n} = \left(\frac{1 + (1 + 2s)^{-1/2}}{2}\right)^n.$$

Markov's inequality and optimization over s give $\mathbb{P}(S_n \leq nc) \leq e^{-nI(c)}$. Moreover,

$$\log \frac{1 + (1 + 2s)^{-1/2}}{2} = -s/2 + O(s^2) \quad (s \downarrow 0),$$

so $I(c) > 0$ for $c < 1/2$. $\blacksquare$

Proof of Theorem 2. For $\alpha < 2$, (33) yields a $\theta > 0$ such that $g_\alpha(\theta) = -\gamma < 0$. The entropy term satisfies

$$LH(1/L) = \log L - (L-1)\log(1-1/L) = \log L + 1 + O(L^{-1}) \leq \log L + 1. \quad (39)$$

The inequality follows from $-\log(1-x) \leq x/(1-x)$. Thus

$$B_L(\alpha, \theta) \leq \log L + 1 + \log 5 + \theta \log 4 - L\gamma < 0$$

for all sufficiently large L , proving the exponential upper bound at every such fixed depth. For $\alpha > 2$, Proposition 11 shows that $\mathbb{P}(K_{n,L}(\alpha) > 1) \rightarrow 1$ at every fixed depth. Hence $\alpha_L \leq 2$ for all L , while every $\alpha < 2$ satisfies $\alpha_L \geq \alpha$ for all sufficiently large L . Consequently $\lim_{L \rightarrow \infty} \alpha_L = 2$.

For the quantitative lower bound, choose $A > 1 + \log 5$ and set

$$\delta_L = \sqrt{\frac{10(\log L + A)}{L}}, \quad \alpha(L) = 2e^{-\delta_L}, \quad \theta_L = \frac{\delta_L}{5}.$$

Equations (33) and (39) give

$$\begin{aligned} B_L(\alpha(L), \theta_L) &= \log L + 1 + \log 5 - \frac{L\delta_L^2}{10} + O(L\delta_L^3) + o(1) \\ &= 1 + \log 5 - A + o(1) < 0, \end{aligned}$$

because $L\delta_L^3 \rightarrow 0$. For each sufficiently large fixed L , Theorem 1 therefore proves that $\alpha(L)$ belongs to the set defining α_L . Thus

$$\alpha_L \geq 2 \exp \left[-\sqrt{\frac{10(\log L + A)}{L}} \right] = 2 \exp \left[-(\sqrt{10} + o(1)) \sqrt{\frac{\log L}{L}} \right].$$

■

Under the coupling $W_\ell^{(\alpha)} = \sqrt{\alpha} W_\ell^{(1)}$, positive homogeneity gives the exact identities

$$F_{n,L}^{(\alpha)} = \alpha^{L/2} F_{n,L}^{(1)}, \quad K_{n,L}(\alpha) = \alpha^{L/2} K_{n,L}(1). \quad (40)$$

The contraction event is therefore monotone in α . The definition of α_L specifies the endpoint of parameters with asymptotically certain nonexpansion; the results above do not assert a complete probability dichotomy at every finite-depth endpoint.

4.4 Independent Symmetric Biases

Theorem 13 (Bias extension) *Suppose all coordinates of $b_1, \dots, b_L \in \mathbb{R}^n$ are mutually independent and symmetric about zero, and the entire bias array is independent of the weights. Let $K_{n,L}^{(b)}(\alpha)$ be the global Lipschitz constant of the network with preactivations $W_\ell x_{\ell-1} + b_\ell$. For $L \geq 2$ and $\theta > 0$,*

$$\mathbb{P}(K_{n,L}^{(b)}(\alpha) > 1) \leq 2e^{nB_L(\alpha, \theta)}. \quad (41)$$

For $\alpha > 2$,

$$\mathbb{P}(K_{n,L}^{(b)}(\alpha) \leq 1) \leq \frac{5\alpha^2 L}{n(\alpha - 2)^2}. \quad (42)$$

No moment assumption on the biases is required.

Proof. For a formal mask, the preactivations have the affine form $H_\ell^D s + c_\ell^D$, where $c_1^D = b_1$ and $c_\ell^D = W_\ell D_{\ell-1} c_{\ell-1}^D + b_\ell$. The strict affine region is now $R_D = \{s : E_\ell^D(H_\ell^D s + c_\ell^D) > 0 \text{ for every } \ell\}$. Their linear parts are unchanged. The conditional Gaussian row argument in Lemma 4 shows that a full-dimensional cell with nonzero Jacobian has strict preactivation signs after subdivision. If a formal linear part vanishes, every subsequent linear part vanishes, so such cells have zero Jacobian. Lemma 3 therefore gives the same reduction with strict affine regions.

Extend (17) by $b_\ell^\Sigma = \Sigma_\ell b_\ell$. The distribution remains invariant and $c_\ell^D(W^\Sigma, b^\Sigma) = \Sigma_\ell c_\ell^D(W, b)$. If c_D stacks these offsets, averaging over the signs now gives, for every nonnegative gauge-invariant functional G ,

$$\mathbb{E}[G \mathbf{1}_{\{R_D \neq \emptyset\}}] = \mathbb{E} \left[G \frac{r(\text{im} A_D + c_D)}{2^{nL}} \right].$$

An affine subspace of dimension at most n in $\mathbb{R}^{nL}$ meets at most $\sum_{j=0}^n \binom{nL}{j}$ open orthants, by the affine arrangement recurrence with boundary values $R(M, 0) = R(0, d) = 1$. For $L \geq 2$, the binomial entropy bound (Cover and Thomas, 2006) yields

$$\sum_{j=0}^n \binom{nL}{j} \leq e^{nLH(1/L)}.$$

Applying the net and radial-moment estimates to the unchanged J_D proves (41), in fact with the leading factor 2 replaced by 1.

For the second claim, couple the biased map F_b and zero-bias map F_0 using the same weights. Put $C_0 = 0$ and $C_\ell = \|W_\ell\|_{2 \rightarrow 2} C_{\ell-1} + \|b_\ell\|$. The 1-Lipschitz property of ReLU gives $\sup_s \|F_b(s) - F_0(s)\| \leq C_L < \infty$ for every realization. Therefore $F_b(cs)/c \rightarrow F_0(s)$ as $c \rightarrow \infty$, by homogeneity of F_0 . For $x \neq y$,

$$K_{n,L}^{(b)}(\alpha) \geq \lim_{c \rightarrow \infty} \frac{\|F_b(cx) - F_b(cy)\|}{c\|x - y\|} = \frac{\|F_0(x) - F_0(y)\|}{\|x - y\|}.$$

Taking the supremum proves $K_{n,L}^{(b)}(\alpha) \geq K_{n,L}(\alpha)$. Now use Proposition 11. This comparison itself holds for arbitrary finite biases, without symmetry or independence. $\blacksquare$

5 Exact Activation-region Counts

Throughout this section the input space is $\mathbb{R}^n$, every layer has width n , and the output layer also applies ReLU. The strict-region count retains different activation patterns even when their output maps coincide. Thus it differs from the number of maximal regions on which the output has a single affine formula, a distinction relevant to the study of activation patterns (Hanin and Rolnick, 2019b). Let $R_{n,L}$ denote this count for the zero-bias network and $R_{n,L}^{(b)}$ its affine counterpart.

5.1 The All-active Pattern

Proposition 14 *For the all-active mask sequence $D_1 = \dots = D_L = I_n$,*

$$\mathbb{P}(R_D \neq \emptyset) = \frac{C_{n,L}}{2^{nL}} = 2^{1-nL} \sum_{j=0}^{n-1} \binom{nL-1}{j}. \quad (43)$$

Proof. Each W_ℓ is invertible almost surely. Set $Q_\ell = W_\ell \cdots W_1$. On $(\text{GL}_n)^L$, the map $(W_1, \dots, W_L) \mapsto (Q_1, \dots, Q_L)$ is a smooth bijection with smooth inverse $W_1 = Q_1$ and $W_\ell = Q_\ell Q_{\ell-1}^{-1}$. Consequently the tuple $(Q_1, \dots, Q_L)$ has a joint density.

Every determinant formed from n rows of the stack of the Q_ℓ is a nonzero polynomial in their entries. To see that it is not identically zero, assign the selected rows distinct standard basis vectors and complete each Q_ℓ to an invertible matrix. The zero set of a nonzero polynomial has Lebesgue measure zero: this follows by induction on the number of variables, viewing the polynomial as a one-variable polynomial with polynomial coefficients and applying Fubini's theorem. A finite union therefore shows that every such minor is nonzero almost surely.

The resulting central arrangement is in general position and has $C_{n,L}$ regions by the central hyperplane-count formula (Cover, 1965). The gauge identity (19), with functional equal to one, proves (43). $\blacksquare$

For fixed $L \geq 2$, the binomial-sum asymptotic gives

$$\log \mathbb{P}(R_D \neq \emptyset) = -n[L \log 2 - LH(1/L)] + o(n), \quad n \rightarrow \infty.$$

Since $LH(1/L) = \log L + 1 + O(L^{-1})$ as $L \rightarrow \infty$, the coefficient is $L \log 2 - O(\log L)$. At depth one, $C_{n,1} = 2^n$ and the probability in (43) is one.

5.2 A Finite Transfer Formula

Put $q = 2^n$ and define

$$B_d(t) = \sum_{j=t}^d \binom{d}{j}, \quad 0 \leq t \leq d.$$

Let $M = M_n$ be the matrix indexed by $0, \dots, n$ with entries

$$M_{r,t} = \begin{cases} 2^{r-t} \binom{n}{r-t} B_{n-r+t}(t), & 0 \leq t \leq r \leq n, \\ 0, & r < t. \end{cases} \quad (44)$$

Write U for its principal submatrix indexed by $1, \dots, n$, and set

$$c_t = \binom{n-1}{t-1} \quad (1 \leq t \leq n), \quad d_t = \binom{n}{t} \quad (0 \leq t \leq n).$$

All entries of these matrices and vectors are nonnegative integers.

Theorem 15 (Exact mean strict-region counts) *For independent Gaussian weights with variance parameter $\alpha > 0$,*

$$\mathbb{E} R_{n,L} = 2q^{1-L} \mathbf{1}^\top U^{L-1} c, \quad L \geq 1. \quad (45)$$

If all bias coordinates are independent centered Gaussian variables of positive variances, independent of the weights, then

$$\mathbb{E} R_{n,L}^{(b)} = q^{1-L} \mathbf{1}^\top M^{L-1} d, \quad L \geq 1. \quad (46)$$

The bias variances may differ across coordinates and layers.

Lemma 16 (Rank of selected formal rows) *Fix intermediate masks $D_1, \dots, D_{L-1}$ and row selections $S_j \subseteq \{1, \dots, n\}$ from the formal preactivation matrices H_j^D . Identify a mask with its retained-coordinate set and write $s_j = |S_j|$. Almost surely the selected stacked rows have rank r_1 , where*

$$r_L = s_L, \quad r_j = s_j + \min\{r_{j+1}, |D_j \setminus S_j|\}, \quad j = L-1, \dots, 1. \quad (47)$$

At any fixed depth this holds simultaneously for all masks and row selections.

Proof. Work backward in coefficient space. Define

$$V_L = \text{span}\{e_i^\top : i \in S_L\}, \quad V_j = \text{span}\{e_i^\top : i \in S_j\} + V_{j+1} W_{j+1} D_j.$$

The subspace V_{j+1} depends only on weights in layers $j+2, \dots, L$ and is independent of W_{j+1} . Conditional on those weights, multiplication of an orthonormal row basis of V_{j+1} by W_{j+1} gives a Gaussian matrix with independent nondegenerate columns. Restricting it to the $|D_j \setminus S_j|$ coordinates outside S_j gives rank $\min\{\dim V_{j+1}, |D_j \setminus S_j|\}$ almost surely. Adding the s_j selected coordinate vectors proves (47). The selected original rows span $V_1 W_1$, whose rank equals $\dim V_1$ because W_1 is invertible almost surely. There are finitely many masks and selections, so the assertion holds simultaneously. This argument does not condition on realizability. $\blacksquare$

Proof of Theorem 15. For fixed intermediate masks, let r_D^0 be the number of open orthants met by the image of the central formal stack. The characteristic-polynomial expansion and the region formula for a central hyperplane arrangement give (Zaslavsky, 1975)

$$r_D^0 = \sum_{s_1, \dots, s_L} (-1)^{s_1 + \dots + s_L - r_1}. \quad (48)$$

This identity remains valid with indexed repeated rows: the alternating sum over all nonempty selections of copies of a fixed nonzero hyperplane is -1 . If a row is zero, the strict orthant count is zero, and the sum in (48) vanishes by pairing selections that differ only in that row.

The formal stack is independent of the final mask. Summing the gauge identity (19) over all masks therefore yields

$$\mathbb{E}R_{n,L} = q^{1-L} \sum_{D_1, \dots, D_{L-1}} \mathbb{E}r_D^0. \quad (49)$$

For a future rank t and a current selection of size s , there are $\binom{n}{s}$ choices of selected coordinates, 2^s mask choices on those coordinates, and $\binom{n-s}{j}$ choices retaining j outside coordinates. The new rank is $s + \min(t, j)$. Thus

$$\mathbb{E}R_{n,L} = q^{1-L} w^\top A^{L-1} v, \quad v_t = (-1)^t \binom{n}{t}, \quad w_r = (-1)^r, \quad (50)$$

where A is indexed by $0, \dots, n$ and

$$A_{r,t} = \sum_{s=0}^n \sum_{j=0}^{n-s} (-2)^s \binom{n}{s} \binom{n-s}{j} \mathbf{1}_{\{r=s+\min(t,j)\}}.$$

To remove the signs, represent a vector by its coefficient polynomial in x , and let $f_t(x)$ represent column t of A . Every $f_t(1)$ is zero, since

$$f_t(1) = \sum_{s=0}^n (-2)^s \binom{n}{s} 2^{n-s} = 0.$$

Moreover, for $1 \leq t \leq n$,

$$f_t(x) - f_{t-1}(x) = (x-1)x^{t-1} \sum_{s=0}^{n-t} (-2)^s \binom{n}{s} B_{n-s}(t)x^s,$$

because $\min(t, j)$ and $\min(t-1, j)$ differ precisely when $j \geq t$. The subspace of polynomials vanishing at one is invariant under A . In its basis $e_t(x) = (x-1)x^{t-1}$, $1 \leq t \leq n$, the matrix entries are

$$(-2)^{r-t} \binom{n}{r-t} B_{n-r+t}(t), \quad 1 \leq t \leq r \leq n.$$

Conjugating by $\text{diag}((-1)^t)$ gives U . The initial polynomial $(1-x)^n$ has e_t -coefficient $(-1)^t c_t$, and $e_t(-1) = 2(-1)^t$. Evaluation at -1 in (50) proves (45).

For the affine case write the formal preactivations as $H_j^D s + a_j$, where

$$a_1 = b_1, \quad a_j = b_j + W_j D_{j-1} a_{j-1}.$$

Conditional on the weights and masks, the linear map from $(b_1, \dots, b_L)$ to $(a_1, \dots, a_L)$ is block lower triangular with identity diagonal. The offsets therefore have a joint density, although they need not be independent.

A selected family of affine hyperplanes has nonempty intersection almost surely exactly when its normal rows are independent. Independent rows define a surjective linear map. For dependent rows, consistent offsets lie in a proper linear subspace and hence occur with probability zero. A zero normal has a nonzero offset almost surely and contributes no hyperplane. The affine characteristic-polynomial and region formulas therefore count independent subsets of the normal rows: every intersecting selected family has rank equal to its cardinality, so its contribution is one (Zaslavsky, 1975).

If the future contains t independent selected rows, choosing s current rows preserves independence of all $t + s$ rows exactly when at least t mask coordinates outside the selection are retained, by Lemma 16. For $0 \leq s \leq n - t$, the number of choices is

$$\binom{n}{s} 2^s B_{n-s}(t) = M_{t+s,t}.$$

No larger s preserves independence. At the last layer the initial counts are $d_t = \binom{n}{t}$. Applying the affine gauge identity and cancelling the q choices of final mask now gives (46). $\blacksquare$

5.3 Fixed-width Depth Asymptotics

Fix n and put $\beta_n = 1 - 2^{-n}$. Write $S_{n,L} = \{F_{n,L}^{(\alpha)} \neq 0\}$ for global survival. Survival and strict-region counts differ: a strict region may have all final preactivations negative and hence have zero output.

Corollary 17 *There are finite positive constants κ_n and σ_n , independent of $\alpha > 0$, such that*

$$\mathbb{E}R_{n,L} \sim \kappa_n \beta_n^L, \quad \mathbb{P}(S_{n,L}) \sim \sigma_n \beta_n^L. \quad (51)$$

Both sequences normalized by β_n^L are nondecreasing. To compute κ_n , set $m = 2^n - 1$, $u_1 = 1$, and

$$u_r = \frac{\sum_{t=1}^{r-1} U_{r,t} u_t}{m - B_n(r)} \quad (2 \leq r \leq n), \quad \kappa_n = \frac{2^{n+1}}{2^n - 1} \sum_{r=1}^n u_r. \quad (52)$$

For $n \geq 2$,

$$\frac{\mathbb{E}R_{n,L}}{\beta_n^L} = \kappa_n + O_n \left(\left[\frac{B_n(2)}{2^n - 1} \right]^L \right). \quad (53)$$

Furthermore,

$$\kappa_n \geq n \beta_n^{-2}, \quad \beta_n^{-1} \leq \sigma_n \leq \kappa_n,$$

and both $L^{-1} \log \mathbb{E}R_{n,L}$ and $L^{-1} \log \mathbb{P}(S_{n,L})$ converge to $\log \beta_n$.

Proof. The diagonal entries of the triangular matrix U are the distinct positive tails $B_n(1), \dots, B_n(n)$. Its largest eigenvalue is $m = B_n(1)$, with positive right eigenvector u given by (52). Every eigenvector belonging to another eigenvalue has first coordinate zero. Since $c_1 = u_1 = 1$, the coefficient of u in the eigenvector decomposition of c is one. Formula (45) proves the mean asymptotic and (53). When $n = 1$, $U = (1)$ and $\kappa_1 = 4$.

Let N_L count strict regions with nonzero final mask. The formal stack does not depend on that mask, so the gauge identity gives $\mathbb{E}N_L = \beta_n \mathbb{E}R_{n,L}$. Conditional on the first L layers, every such region has output $x_L(s) = As$ with $A \neq 0$, since at least one output coordinate is strictly positive there. For a fresh Gaussian row w , $wA \neq 0$ almost surely. Removing the next layer's proper hyperplanes leaves at least one strict open subregion in each region counted by N_L . Different old patterns remain distinct after extension, and consequently $R_{n,L+1} \geq N_L$ almost surely. Taking expectations proves monotonicity of $\mathbb{E}R_{n,L}/\beta_n^L$.

Couple all depths by one infinite independent sequence of weight matrices and put $p_L = \mathbb{P}(S_{n,L})$. On $S_{n,L}$, select the first rational input in a fixed enumeration of $\mathbb{Q}^n$ with nonzero output. Continuity ensures its existence, and the selection is measurable with respect to the first L layers. Conditional on these layers, the next layer preserves this output with probability exactly β_n ; hence $p_{L+1} \geq \beta_n p_L$.

Almost surely, a nonzero network has a nonempty strict region with nonzero output. Indeed, on any open set with nonzero output all preceding layer outputs are nonzero, and each fresh Gaussian preactivation has a proper zero hyperplane on each corresponding nonzero linear piece; successively removing these finitely many hyperplanes leaves a strict point. Thus $p_L \leq \mathbb{E}R_{n,L}$. The sequence p_L/β_n^L is nondecreasing and bounded by κ_n , so it has a finite limit σ_n . Since $p_1 = 1$, this limit is at least β_n^{-1} . The lower bound on κ_n follows from (54) below. Positive rescaling of the weights preserves all signs and zero-output events, proving independence from α . ■

The following bounds hold at finite depth:

$$n\beta_n^{L-2} \leq \mathbb{E}R_{n,L} \leq C_{n,L}\beta_n^{L-1}, \quad L \geq 2, \quad (54)$$

$$\beta_n^{L-1} \leq \mathbb{P}(S_{n,L}) \leq \min\{1, C_{n,L}\beta_n^L\}, \quad L \geq 1. \quad (55)$$

For the mean lower bound, retain intermediate sequences whose first mask has rank one and whose remaining masks are nonzero. There are $n(q-1)^{L-2}$ such sequences. Every formal row after the first layer is almost surely a nonzero multiple of the same row of W_1 . Their central arrangement therefore has exactly q regions, and (49) gives the lower bound. A zero intermediate mask kills the strict orthant count. The remaining $(q-1)^{L-1}$ sequences have at most $C_{n,L}$ regions each, giving the mean upper bound.

For survival, apply the gauge identity to the invariant functional $\mathbf{1}_{\{J_D \neq 0\}}$ and bound the survival indicator by the number of realized patterns with nonzero Jacobian. Any zero mask, including the last, forces $J_D = 0$, leaving at most $(q-1)^L$ contributing sequences; the central count is at most $C_{n,L}$. This gives the upper bound in (55). For the lower bound, choose $s = W_1^{-1}e_1$, whose first-layer output is e_1 , and propagate this nonzero vector through the remaining independent layers.

At width one,

$$\mathbb{E}R_{1,L} = 2^{2-L}, \quad \mathbb{P}(S_{1,L}) = 2^{1-L}, \quad (\kappa_1, \sigma_1) = (4, 2).$$

The next mean constants are $\kappa_2 = 8$ and $\kappa_3 = 240/7$. Corollary 17 gives neither an explicit higher-width survival constant nor a rate of convergence for the survival ratio. All of its limits keep the width fixed.

5.4 The Limiting Mean with Independent Biases

Corollary 18 *Under the bias assumptions of Theorem 15, define $z_0 = 1$ and*

$$z_r = \frac{\sum_{t=0}^{r-1} M_{r,t} z_t}{2^n - B_n(r)} \quad (1 \leq r \leq n), \quad \rho_n = \sum_{r=0}^n z_r. \quad (56)$$

Then ρ_n is a finite positive rational number and

$$\mathbb{E}R_{n,L}^{(b)} = \rho_n + O_n(\beta_n^L). \quad (57)$$

On one infinite independent-layer sequence, the geometric open strict cells eventually stop changing almost surely. Their eventual number has expectation ρ_n .

Proof. The eigenvalues of M are the distinct tails $B_n(0), \dots, B_n(n)$. Its largest eigenvalue is $q = 2^n$, with right eigenvector z in (56). Since $d_0 = z_0 = 1$, the coefficient of z in the eigenvector decomposition of d is one. All other eigenvalues are at most $q - 1$, and (46) proves (57). The recursion also proves positivity and rationality of ρ_n .

On each old strict cell, a new preactivation is almost surely either nonconstant affine or a nonzero constant. The latter assertion follows from the fresh nondegenerate bias, conditional on all preceding layers and the new weights. A nonconstant preactivation can only split an old cell by a proper affine hyperplane. Thus the strict-cell counts are nondecreasing, and any change of the geometric open cells strictly increases their number. Monotone convergence and (57) give a finite integer limiting count almost surely, with expectation ρ_n . Only finitely many refinements can therefore occur. ■

The first limits are $\rho_1 = 3$, $\rho_2 = 21$, and $\rho_3 = 279$. There is also a direct qualitative stabilization argument. After the first layer, an independent layer whose weights and biases are all negative maps the entire nonnegative image to zero. This event has probability 2^{-n^2-n} at each depth and therefore occurs almost surely. Subsequent outputs are independent of the input. The corollary additionally gives the exact finite expected number of eventual cells. The cells stabilize as subsets of input space; their depth-indexed labels and constant output values may continue to change. Formula (46) requires nondegenerate biases and does not extend to zero bias variance by substitution.

5.5 Width Two

Proposition 19 For every $L \geq 1$ and $\alpha > 0$,

$$\mathbb{E}R_{2,L} = 8 \left[\left(\frac{3}{4} \right)^L - \left(\frac{1}{4} \right)^L \right], \quad (58)$$

$$\mathbb{E}R_{2,L}^{(b)} = 21 - 24 \left(\frac{3}{4} \right)^L + 4 \left(\frac{1}{4} \right)^L, \quad (59)$$

where the second identity assumes the biases in Theorem 15.

Proof. For $n = 2$ the matrices and initial vectors are

$$U = \begin{pmatrix} 3 & 0 \\ 4 & 1 \end{pmatrix}, \quad M = \begin{pmatrix} 4 & 0 & 0 \\ 8 & 3 & 0 \\ 4 & 4 & 1 \end{pmatrix}, \quad c = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad d = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

Evaluating their powers in (45) and (46) gives the stated identities, including depth one, when both means equal four. ■

The first six zero-bias means are 4, 4, 3.25, 2.5, 1.890625, 1.421875. Their decay concerns strict activation regions; a network that is identically zero still has a linear output map on its whole input space. With nondegenerate biases, the strict partition instead refines and its expected count increases to 21.

6 Finite-horizon Dynamics with Tied Weights

Consider the iteration

$$x_{t+1} = \phi(Wx_t), \quad W \in \mathbb{R}^{N \times N}, \quad W_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, N^{-1}), \quad x_0 \in S^{N-1}, \quad (60)$$

where the same matrix is used at every step. Matrix reuse makes the state x_t dependent on W . Gaussian conditioning for adaptively generated subspaces provides a way to retain the independent Gaussian components needed for concentration. Related conditioning arguments underlie the analysis of approximate message passing and tensor programs (Bayati and Montanari, 2011; Yang, 2019). The result below gives an exponential deviation bound for a fixed number of ReLU iterations, with constants uniform in the deterministic starting vector.

Theorem 20 (Finite-horizon norm concentration) *For every integer $T \geq 0$ and every $\delta > 0$, there are constants $C, c > 0$, depending only on T and δ , such that, for every $N \geq 1$ and every deterministic $x_0 \in S^{N-1}$,*

$$\mathbb{P}_W \left(\max_{0 \leq t \leq T} |2^t \|x_t\|^2 - 1| > \delta \right) \leq Ce^{-cN}. \quad (61)$$

The same bound holds jointly over W and a random unit starting vector independent of W .

For a fixed matrix, define its set of exceptional starting vectors by

$$\mathcal{B}(W) = \left\{ x_0 \in S^{N-1} : \max_{0 \leq t \leq T} |2^t \|x_t\|^2 - 1| > \delta \right\}.$$

Corollary 21 (Exceptional starting vectors) *Let μ_N be a deterministic probability measure on S^{N-1} , or a random probability measure independent of W . With the constants from Theorem 20, conditionally on μ_N ,*

$$\mathbb{P}_W \left(\mu_N(\mathcal{B}(W)) > e^{-cN/2} \right) \leq Ce^{-cN/2}. \quad (62)$$

The horizon T is fixed in these statements. Uniformity of the constants in a deterministic x_0 does not replace the probability in (61) by a probability of simultaneous control over all starting vectors. Corollary 21 supplies a simultaneous statement relative to a prescribed input measure.

6.1 Gaussian Conditioning Under Adaptive Queries

An orthogonal query Wu reveals the component of each row of W in the direction u . Conditional on previously revealed components, the remaining components remain Gaussian. The following formulation allows the query directions and their orthogonal complements to be random.

Lemma 22 (Adaptive Gaussian innovations) *Let W have independent $\mathcal{N}(0, N^{-1})$ entries, let $\mathcal{F}_0$ be a sigma-field independent of W , and fix $K \leq N$. For $1 \leq k \leq K$, let u_k be an $\mathcal{F}_{k-1}$ -measurable unit vector orthogonal to $u_1, \dots, u_{k-1}$, and set*

$$g_k = Wu_k, \quad \mathcal{F}_k = \mathcal{F}_0 \vee \sigma(g_1, \dots, g_k), \quad P_k = \sum_{j=1}^k u_j u_j^\top, \quad M_k = \sum_{j=1}^k g_j u_j^\top.$$

Then, conditionally on $\mathcal{F}_k$,

$$W \stackrel{d}{=} M_k + Z(I - P_k), \quad (63)$$

where Z is an independent copy of W . More precisely, for every bounded Borel function Ψ ,

$$\mathbb{E}[\Psi(W) \mid \mathcal{F}_k] = \mathbb{E}_Z[\Psi(M_k + Z(I - P_k))],$$

with M_k, P_k fixed inside the expectation over Z . Furthermore, $g_1, \dots, g_K$ are independent $\mathcal{N}(0, I_N/N)$ vectors and are jointly independent of $\mathcal{F}_0$.

Proof. First fix a deterministic orthogonal projection P and a deterministic unit vector u with $Pu = 0$. Put $P^+ = P + uu^\top$. For a fresh Gaussian matrix Z ,

$$Z(I - P) = (Zu)u^\top + Z(I - P^+).$$

The vector $\xi = Zu$ has law $\mathcal{N}(0, I_N/N)$. For a row vector $z^\top$ of Z , the covariance between $z^\top u$ and $(I - P^+)z$ is

$$(I - P^+)\mathbb{E}[zz^\top]u = N^{-1}(I - P^+)u = 0.$$

Joint Gaussianity and independence of the rows show that ξ and $R = Z(I - P^+)$ are independent. The residual R has the law of $\tilde{Z}(I - P^+)$ for a fresh Gaussian matrix $\tilde{Z}$.

To prove (63) by induction, set $M_0 = P_0 = 0$. The assertion at $k = 0$ follows from independence of W and $\mathcal{F}_0$. Given the assertion at $k - 1$, condition on $\mathcal{F}_{k-1}$. All previous query vectors and u_k are then fixed. Since $M_{k-1}u_k = 0$ and $(I - P_{k-1})u_k = u_k$, the conditional law of g_k is that of Zu_k , namely $\mathcal{N}(0, I_N/N)$. Apply the deterministic decomposition above with $P = P_{k-1}$ and $u = u_k$: the conditional joint law of (g_k, W) is the law of

$$\left(\xi, M_{k-1} + \xi u_k^\top + R \right),$$

where ξ and R are independent. Further conditioning on g_k therefore leaves a fresh residual with law $Z(I - P_k)$ and gives (63). This proves the displayed conditional expectation identity as well. The conditional law of each g_k is constant, independently of $\mathcal{F}_{k-1}$; iteration of conditional expectations proves joint independence of the innovations and their independence of $\mathcal{F}_0$. $\blacksquare$

The argument requires orthogonality, measurability with respect to past queries, and independent Gaussian rows; it does not apply as stated to a symmetric Gaussian matrix.

6.2 A Finite-dimensional Representation of the Trajectory

Suppose $N \geq T + 1$. Apply Gram–Schmidt to $x_0, \dots, x_T$ in chronological order. Start with $v_1 = x_0$ and query $g_1 = Wv_1$. Whenever a state has a nonzero residual orthogonal to the previously queried span, normalize that residual to obtain the next vector v_j and query $g_j = Wv_j$. If a state is already in the span, its image under W is determined by the previous answers, so the iteration can continue without a new query. In particular, every new query direction is a measurable function of the previous answers.

After processing through x_T , extend the query list to exactly $d = T + 1$ orthonormal vectors. For this extension, repeatedly project the first standard basis vector with nonzero residual onto the orthogonal complement of the current span and normalize. This is a measurable rule based on the

existing transcript. It makes the total number of queries deterministic without conditioning on the random number of trajectory directions. The additional vectors are orthogonal to $\text{span}(x_0, \dots, x_T)$.

Lemma 22 gives independent $g_1, \dots, g_d \sim \mathcal{N}(0, I_N/N)$. Define

$$a_t = (\langle x_t, v_1 \rangle, \dots, \langle x_t, v_d \rangle), \quad \Gamma_i = \sqrt{N}((g_1)_i, \dots, (g_d)_i).$$

Then $\Gamma_1, \dots, \Gamma_N$ are independent $\mathcal{N}(0, I_d)$ vectors. Parseval's identity and linearity give

$$\|a_t\| = \|x_t\|, \quad Wx_t = \sum_{j=1}^d (a_t)_j g_j, \quad \|x_{t+1}\|^2 = \frac{1}{N} \sum_{i=1}^N (a_t \cdot \Gamma_i)_+^2. \quad (64)$$

The coefficients a_t depend on the same Gaussian vectors appearing in the last expression. Consequently, the required concentration estimate must be uniform over the coefficient vector.

6.3 Concentration Estimates

The concentration and covering estimates below follow the methods in Vershynin (2018).

Lemma 23 (Bernstein bound) *Let $X_1, \dots, X_N$ be independent mean-zero random variables with $\mathbb{E}e^{|X_i|/K} \leq 2$ for some $K > 0$. For every $u > 0$,*

$$\mathbb{P}\left(\left|N^{-1} \sum_{i=1}^N X_i\right| > u\right) \leq 2 \exp\left[-N \min\left\{\frac{u^2}{16K^2}, \frac{u}{8K}\right\}\right].$$

Proof. Expansion of the exponential gives $\mathbb{E}|X_i|^p \leq 2p!K^p$. Hence, for $|\lambda|K \leq 1/4$,

$$\mathbb{E}e^{\lambda X_i} \leq 1 + 2 \sum_{p \geq 2} (|\lambda|K)^p \leq e^{4\lambda^2 K^2}.$$

Independence and Markov's inequality bound the upper tail by $\exp(-\lambda Nu + 4N\lambda^2 K^2)$. For $u \leq 2K$, take $\lambda = u/(8K^2)$; for $u > 2K$, take $\lambda = 1/(4K)$. These choices give, respectively, the exponents $-Nu^2/(16K^2)$ and at most $-Nu/(8K)$. Apply the same argument to $-X_i$ for the lower tail. ■

Lemma 24 (Uniform concentration in a fixed dimension) *Fix an integer $d \geq 1$, a radius $R \geq 0$, and $\varepsilon > 0$. For independent $\Gamma_1, \dots, \Gamma_N \sim \mathcal{N}(0, I_d)$, there are $C, c > 0$, depending only on d, R, ε , such that*

$$\mathbb{P}\left(\sup_{\|a\| \leq R} \left|\frac{1}{N} \sum_{i=1}^N (a \cdot \Gamma_i)_+^2 - \frac{1}{2} \|a\|^2\right| > \varepsilon\right) \leq Ce^{-cN}. \quad (65)$$

Proof. For $R = 0$, the supremum is zero. Suppose $R > 0$, and put $A_N(a) = N^{-1} \sum_i (a \cdot \Gamma_i)_+^2$. For fixed a with $\|a\| \leq R$, symmetry of the normal distribution gives $\mathbb{E}A_N(a) = \|a\|^2/2$. Write $Y_i = (a \cdot \Gamma_i)_+^2$ and $X_i = Y_i - \mathbb{E}Y_i$. Setting $Z_i = a \cdot \Gamma_i$ gives $|X_i| \leq Z_i^2 + R^2/2$. Taking $K_R = 32R^2$ and using the Gaussian square moment-generating function,

$$\mathbb{E}e^{|X_i|/K_R} \leq e^{1/64}(1 - 1/16)^{-1/2} < 2.$$

Lemma 23, with $u = \varepsilon/2$, therefore gives

$$\mathbb{P}(|A_N(a) - \|a\|^2/2| > \varepsilon/2) \leq 2e^{-c_1 N}, \quad c_1 = \min \left\{ \frac{\varepsilon^2}{64K_R^2}, \frac{\varepsilon}{16K_R} \right\}. \quad (66)$$

For $\|a\|, \|b\| \leq R$, the Lipschitz property of ReLU implies

$$|(a \cdot \Gamma)_+^2 - (b \cdot \Gamma)_+^2| \leq 2R\|a - b\|\|\Gamma\|^2.$$

Also $|\|a\|^2/2 - \|b\|^2/2| \leq R\|a - b\|$. Set

$$h = \min \left\{ R, \frac{\varepsilon}{2R(4d+1)} \right\}.$$

A maximal h -separated subset of the radius- R ball is an h -net with at most $(1 + 2R/h)^d \leq (3R/h)^d$ points, by disjoint-ball volume comparison. Its size is independent of N . Finally, since $\sum_i \|\Gamma_i\|^2$ has a chi-square law with dN degrees of freedom, Markov's inequality at exponential parameter $1/4$ gives

$$\mathbb{P} \left(N^{-1} \sum_i \|\Gamma_i\|^2 > 2d \right) \leq \exp \left[-\frac{1 - \log 2}{2} dN \right]. \quad (67)$$

On the complementary event, and when (66) holds at every net point, approximate any a by a net point b . The total error is at most

$$4Rdh + \varepsilon/2 + Rh \leq \varepsilon.$$

A union bound over the net, together with (67), proves (65). ■

Lemma 25 (An operator-norm estimate) *For W as in (60),*

$$\mathbb{P}(\|W\|_{2 \rightarrow 2} > 8) \leq 2 \exp[-(8 - \log 81)N].$$

Proof. A $1/4$ -net $\mathcal{N} \subset S^{N-1}$ can be chosen with $|\mathcal{N}| \leq 9^N$. Approximating each of two unit vectors by net points gives, for every matrix A ,

$$\|A\|_{2 \rightarrow 2} \leq 2 \max_{u, v \in \mathcal{N}} |u^\top A v|.$$

For fixed unit vectors u, v , the variable $u^\top W v$ is centered Gaussian with variance $1/N$, so $\mathbb{P}(|u^\top W v| > 4) \leq 2e^{-8N}$. A union bound over at most 81^N pairs proves the claim. ■

6.4 Proof of Finite-horizon Concentration

Proof of Theorem 20. The assertion is immediate for $T = 0$. Let $T \geq 1$ and first assume $N \geq T + 1$. On the event $\mathcal{O} = \{\|W\|_{2 \rightarrow 2} \leq 8\}$,

$$\|x_t\| \leq 8^t \leq R := 8^T \quad (0 \leq t \leq T).$$

Use the $d = T + 1$ queries from Section 6.2 and apply Lemma 24 with radius R and tolerance $\eta = \delta 2^{-(T+2)}$. Let $\mathcal{E}$ denote the event on which its uniform bound holds. Then $\mathbb{P}(\mathcal{E}^c) \leq C_1 e^{-c_1 N}$.

On $\mathcal{O} \cap \mathcal{E}$, every coefficient a_t in (64) belongs to the radius- R ball, and uniformity permits its substitution even though it depends on the innovations. Consequently,

$$|\|x_{t+1}\|^2 - \frac{1}{2}\|x_t\|^2| \leq \eta \quad (0 \leq t < T).$$

The errors $e_t = |\|x_t\|^2 - 2^{-t}|$ satisfy $e_0 = 0$ and $e_{t+1} \leq \eta + e_t/2$, whence $e_t \leq 2\eta$. Thus

$$\max_{t \leq T} |2^t \|x_t\|^2 - 1| \leq 2^{T+1} \eta = \delta/2 < \delta.$$

Lemma 25 and the bound on $\mathbb{P}(\mathcal{E}^c)$ prove (61) for $N \geq T + 1$. Increasing C covers the finitely many smaller dimensions, since their probabilities are at most one. For a random start independent of W , condition on its value and integrate the deterministic bound, whose constants are uniform in that value. $\blacksquare$

Proof of Corollary 21. For every fixed t , the state x_t is a continuous function of (x_0, W) . The exceptional-input indicator is therefore Borel measurable. For deterministic μ_N , Tonelli's theorem and Theorem 20 imply

$$\mathbb{E}_W \mu_N(\mathcal{B}(W)) = \int \mathbb{P}_W(x_0 \in \mathcal{B}(W)) \mu_N(dx_0) \leq C e^{-cN}.$$

Markov's inequality at threshold $e^{-cN/2}$ gives (62). If μ_N is random and independent of W , condition on μ_N and apply the same argument. $\blacksquare$

6.5 A Reused Gaussian Bias

Now let

$$x_{t+1} = \phi(Wx_t + b), \quad b_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2/N),$$

where b is independent of W and is also reused at each step. Define

$$\bar{q}_0 = 1, \quad \bar{q}_{t+1} = \frac{1}{2}(\bar{q}_t + \sigma^2), \quad \bar{q}_t = \sigma^2 + (1 - \sigma^2)2^{-t}. \quad (68)$$

Theorem 26 (A reused Gaussian bias) *For every integer $T \geq 0$, every $\delta > 0$, and every $\sigma \geq 0$, there are $C, c > 0$, depending only on these parameters, such that for every $N \geq 1$ and every deterministic $x_0 \in S^{N-1}$,*

$$\mathbb{P}_{W,b} \left(\max_{0 \leq t \leq T} |\|x_t\|^2 - \bar{q}_t| > \delta \right) \leq C e^{-cN}.$$

The independent-random-start conclusion of Theorem 20 and the exceptional-measure conclusion of Corollary 21 also hold, with independence now required from (W, b) .

Proof. The cases $T = 0$ and $\sigma = 0$ follow immediately from the definitions and Theorem 20. Assume $T \geq 1$, $\sigma > 0$, and $N \geq T + 1$. Apply Lemma 22 with $\mathcal{F}_0 = \sigma(b)$ to the Gram–Schmidt queries of the biased trajectory. The resulting $d = T + 1$ Gaussian innovations are jointly independent of b . With the notation of (64), put

$$\Gamma'_i = (\Gamma_i, \sqrt{N}b_i/\sigma), \quad a'_i = (a_i, \sigma).$$

The vectors Γ'_i are independent $\mathcal{N}(0, I_{d+1})$, and

$$\|x_{t+1}\|^2 = \frac{1}{N} \sum_i (a'_t \cdot \Gamma'_i)_+^2, \quad \|a'_t\|^2 = \|x_t\|^2 + \sigma^2. \quad (69)$$

On $\{\|W\|_{2 \rightarrow 2} \leq 8, \|b\|^2 \leq 2\sigma^2\}$, the inequality $\|x_{t+1}\| \leq 8\|x_t\| + 2\sigma$ yields

$$\|a'_t\| \leq R' := 8^T (1 + 3\sigma) \quad (0 \leq t \leq T).$$

The complementary probability is at most Ce^{-cN} by Lemma 25 and (67) in dimension one. Apply Lemma 24 in dimension $d+1$, with radius R' and tolerance $\eta = \delta/4$. Except on another event of probability at most Ce^{-cN} , (69) then gives

$$|\|x_{t+1}\|^2 - \frac{1}{2}(\|x_t\|^2 + \sigma^2)| \leq \eta \quad (0 \leq t < T).$$

For $e_t = |\|x_t\|^2 - \bar{q}_t|$, this implies $e_0 = 0$, $e_{t+1} \leq \eta + e_t/2$, and hence $e_t \leq 2\eta < \delta$. Increasing C covers $N < T+1$. Conditioning on an independent start proves the random-start statement; Tonelli's theorem and Markov's inequality prove the measure statement as in Corollary 21. $\blacksquare$

7 Numerical Evaluation

The numerical results evaluate the finite-width bounds and exact counting formulas, and compare the norm identities with Monte Carlo simulations. The script `code/reproduce.py` generates all five tables. Each simulation configuration has a recorded pseudorandom seed.

7.1 Sufficient Thresholds and Probability Bounds

Define the sufficient threshold

$$\alpha_L^* = \sup\{\alpha > 0 : \inf_{\theta > 0} B_L(\alpha, \theta) < 0\}.$$

Thus $\alpha_L^* \leq \alpha_L$. Table 1 evaluates this quantity using g_α , and compares it with the leading large-depth expression. The stationary equations are solved by bracketing a scalar root; an independent scalar minimization checks representative values. The computed values are sufficient thresholds; the true thresholds α_L may be larger.

The first sufficient depths obtained from this calculation are 119, 620, 3864, 50026, 231065, and 7314489 for $\alpha = 1, 1.4, 1.7, 1.9, 1.95, 1.99$, respectively. The script evaluates the optimized bound at both adjacent integer depths. At $\alpha = 1$ and $L = 1000$, $c_L(1) \approx 54.81388684686$. Rounding the rate downward gives $\mathbb{P}(K_{n,L}(1) > 1) \leq 2\exp(-54.81388684n)$.

For $\alpha > 2$, Table 2 compares the two upper bounds on $\mathbb{P}(K_{n,L}(\alpha) \leq 1)$. The Chernoff rate is evaluated at its unique stationary point. Writing $z = (1 + 2s)^{-1/2}$ reduces its stationary equation to $z^3/(1+z) = 1/\alpha$. Entries exceeding one are retained to show where the untruncated bounds are uninformative.

7.2 Exact Region Means

Table 3 evaluates the finite transfer matrices using integer arithmetic and rational division. It contrasts the zero-bias strict-region count with the count for independent centered nondegenerate Gaussian

L	α_L^*	$2 \exp(-\sqrt{10 \log L/L})$	$c_L(1)$
10	0.3670	0.4386	—
30	0.6298	0.6896	—
100	0.9537	1.0146	—
300	1.2369	1.2932	10.8
1,000	1.4940	1.5378	54.8
3,000	1.6679	1.6986	183
10,000	1.7980	1.8170	634
100,000	1.9268	1.9333	6.45×10^3
1,000,000	1.9746	1.9766	6.46×10^4

Table 1: Sufficient thresholds and the corresponding width rates $c_L(1)$. A dash denotes a nonpositive rate. The leading asymptotic expression is included for comparison and is not itself a finite-depth certificate.

α	$I(1/\alpha)$	$5\alpha^2 L/[n(\alpha - 2)^2]$	$Le^{-nI(1/\alpha)}$
2.2	0.0008735	6.05	4.17
2.5	0.0045451	1.25	0.106
3.0	0.0139492	0.45	8.75×10^{-6}
4.0	0.0364211	0.2	1.52×10^{-15}
6.0	0.0784667	0.113	8.36×10^{-34}

Table 2: Upper bounds on the nonexpansiveness probability at $L = 10$ and $n = 1000$. All entries are evaluations of analytic bounds.

biases. The values do not depend on the positive weight variance parameter or on the positive bias variances. They count strict activation patterns, including distinct patterns whose final output maps agree.

At width two, the transfer calculation agrees exactly, through depth 25, with $8[(3/4)^L - (1/4)^L]$ and $21 - 24(3/4)^L + 4(1/4)^L$. For $n = 1, 2, 3$, the limiting constants are $\kappa_n = 4, 8, 240/7$ and $\rho_n = 3, 21, 279$, respectively.

7.3 Radial Moments

For a fixed formal mask sequence D , Table 4 compares $\mathbb{E}\|J_D e_1\|^{2p}$ with simulations of the full independent Gaussian matrices. At $n = 4$, the masks are 1101, 1001, 1110; at $n = 6$, they are 101101, 011011, with the binary strings denoting the mask diagonals. The remaining rows sample $Y_n = (\alpha/n) \sum_{i=1}^n B_i Z_i^2$, with independent $B_i \sim \text{Bernoulli}(1/2)$ and $Z_i \sim N(0, 1)$, and compare $\mathbb{E}Y_n^p$ with the binomial–chi-square mixture formula.

The discrepancies in these six rows range from 0.09 to 2.01 estimated standard errors in absolute value. The recorded seeds are 202609071 and 202609072 for the two fixed-mask configurations and 202609073 and 202609074 for the two mixture configurations.

n	L	$\mathbb{E}R_{n,L}$	$\mathbb{E}R_{n,L}^{(b)}$
2	1	4	4
2	2	4	7.75
2	3	3.25	10.9375
2	4	2.5	13.4219
2	5	1.89062	15.3086
2	6	1.42188	16.7295
3	2	14.5	25.75
3	3	17	49.0625
4	2	52	88.1875
4	3	86.6875	231.684

Table 3: Exact transfer-matrix means, displayed to at most six significant digits. The output files also retain their exact rational values.

Variable	n	L	α	p	Exact mean	Sample mean	SE
$\ J_{De_1}\ ^{2p}$	4	3	1.3	1	0.61791	0.62016	0.00299
$\ J_{De_1}\ ^{2p}$	4	3	1.3	2	2.1212	2.1682	0.0681
$\ J_{De_1}\ ^{2p}$	6	2	1.3	1.5	0.89871	0.90369	0.004
Y_n^p	6	1	1.3	1	0.65	0.65267	0.00133
Y_n^p	6	1	1.3	2.5	0.96763	0.97826	0.00568
Y_n^p	12	1	2	1.7	1.2401	1.2398	0.00307

Table 4: Moment comparisons using 200000 independent samples per configuration. SE is the sample standard deviation of the displayed variable divided by $\sqrt{200000}$. Rows with identical masks or identical mixture parameters share samples across powers.

7.4 Tied and Independent Layers

For each width, start from the deterministic unit vector $x_0 = e_1$ and record

$$\Delta_{N,8} = \max_{0 \leq t \leq 8} |2^t \|x_t\|^2 - 1|.$$

The tied experiment uses one matrix with independent $N(0, 1/N)$ entries at all eight steps. The independent-layer experiment draws a fresh matrix with the same entry distribution at each step. Each column uses 300 independently drawn trajectories, with different seeds for the two experiments.

Both sets of sample medians decrease with width in this experiment. Tied layers have larger upper quantiles at the three widths considered.

8 Discussion

The global stability threshold approaches 2 when width tends to infinity before depth. The finite-width certificate depends on an arrangement bound and a sphere-net estimate, both of which lose

N	Tied layers			Independent layers		
	Median	90% quantile	$\hat{p} \pm \text{SE}$	Median	90% quantile	$\hat{p} \pm \text{SE}$
50	0.844	4.716	0.837 ± 0.021	0.721	1.694	0.850 ± 0.021
100	0.689	2.570	0.673 ± 0.027	0.544	0.954	0.563 ± 0.029
300	0.467	1.311	0.457 ± 0.029	0.337	0.618	0.207 ± 0.023

Table 5: Relative norm deviations over eight steps. The probability entries use $\hat{p} = \widehat{\mathbb{P}}(\Delta_{N,8} > 1/2)$ and give sample proportions plus or minus one estimated standard error, $\sqrt{\hat{p}(1-\hat{p})/300}$; they are not confidence intervals. Quantiles are descriptive sample statistics.

information about individual activation regions. The scalar moment exponent is exact; sharper finite-depth thresholds require better control of the arrangement and sphere-net steps.

At fixed width, the transfer matrices give exact mean strict-region counts and their depth asymptotics. Nonzero strict trajectories disappear in the homogeneous network, while the activation partition stabilizes in the biased network. These conclusions concern a separate order of limits. For tied weights, the adaptive-query coefficient space has dimension at most $T + 1$. Bounds for matrix-dependent inputs or horizons growing with width require additional uniform control beyond the fixed-dimensional argument.

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Code Availability. The accompanying source archive contains the Python code, fixed-seed numerical outputs, table generators, and reproduction instructions for Section 7.

Appendix A. Exact Scalar Moment Exponent

The scalar bound in Lemma 10 has the exact exponential rate as $n \rightarrow \infty$. For $\theta > 0$ and $u \in [0, 1]$, define

$$\Phi_{\alpha,\theta}(u) = H(u) - \log 2 + \theta \log(2\alpha) + \left(\frac{u}{2} + \theta\right) \log\left(\frac{u}{2} + \theta\right) - \frac{u}{2} \log \frac{u}{2} - \theta. \quad (70)$$

This function is continuous on $[0, 1]$ under the convention $0 \log 0 = 0$.

Proposition 27 (Exact high-moment exponent) *For every $\alpha > 0$ and $\theta > 0$,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}[(X_n^{(\alpha)})^{\theta n}] = \max_{0 \leq u \leq 1} \Phi_{\alpha, \theta}(u) = g_\alpha(\theta). \quad (71)$$

The maximum is attained uniquely at $u_\theta = 1/(1 + z_\theta)$, with z_θ from (30).

Proof. For $m \geq 1$ and $r > 0$, integration against the chi-square density gives $\mathbb{E}[(\chi_m^2)^r] = 2^r \Gamma(m/2 + r) / \Gamma(m/2)$. Consequently,

$$\mathbb{E}[(X_n^{(\alpha)})^{\theta n}] = 2^{-n} \sum_{m=1}^n \binom{n}{m} \left(\frac{2\alpha}{n} \right)^{\theta n} \frac{\Gamma(m/2 + \theta n)}{\Gamma(m/2)}. \quad (72)$$

The term with $m = 0$ vanishes. Uniformly for $1 \leq m \leq n$, with $u = m/n$, Stirling's estimates give

$$\begin{aligned} \log \binom{n}{m} &= nH(u) + O(\log n), \\ \log \frac{\Gamma(m/2 + \theta n)}{\Gamma(m/2)} &= \theta n \log n + n \left[\left(\frac{u}{2} + \theta \right) \log \left(\frac{u}{2} + \theta \right) - \frac{u}{2} \log \frac{u}{2} - \theta \right] + O_\theta(\log n). \end{aligned}$$

For the second estimate, one may use $\log \Gamma(x) = (x - 1/2) \log x - x + O(1)$ uniformly on $x \geq 1/2$; the remaining logarithmic ratio is $O_\theta(\log n)$ even when $m = 1$. If $a_{n,m}$ is the m th summand in (72), then

$$\frac{1}{n} \log a_{n,m} = \Phi_{\alpha, \theta}(m/n) + O_{\alpha, \theta}(\log n/n)$$

uniformly. A sum of n nonnegative terms lies between its largest term and n times that term. Continuity of Φ and density of the grids $\{1/n, \dots, 1\}$ therefore prove the variational limit.

For $0 < u < 1$,

$$\begin{aligned} \partial_u \Phi_{\alpha, \theta} &= \log \frac{1-u}{u} + \frac{1}{2} \log \frac{u/2 + \theta}{u/2}, \\ \partial_{uu} \Phi_{\alpha, \theta} &= -\frac{1}{u(1-u)} + \frac{1}{4(u/2 + \theta)} - \frac{1}{2u} < 0. \end{aligned}$$

The first derivative tends to $+\infty$ at zero and $-\infty$ at one, so there is a unique interior maximizer. Its stationary equation, after substitution $u = 1/(1 + z)$, is $\theta = (1 - z)/(2z^2)$. Thus $z = z_\theta$ and

$$\frac{u_\theta}{2} + \theta = \frac{1}{2z_\theta^2(1 + z_\theta)}.$$

Substitution into (70) gives exactly $g_\alpha(\theta)$ in (30). ■

In particular, $u_\theta = 1/2 + \theta/2 + O(\theta^2)$ as $\theta \downarrow 0$. The exact scalar rate does not make the global Lipschitz bound sharp: the arrangement comparison and sphere-net estimate in Proposition 9 remain upper bounds. Accordingly, $\sqrt{10}$ in Theorem 2 is a constant in a certified lower bound for α_L .

Lemma 28 (Exponential rate of the arrangement bound) *For every fixed $L \geq 2$,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log C_{n,L} = LH(1/L).$$

Proof. For $L > 2$, the binomial coefficients in (20) increase through $j = n - 1$, so

$$\binom{nL-1}{n-1} \leq \frac{C_{n,L}}{2} \leq n \binom{nL-1}{n-1}.$$

Stirling’s formula gives the asserted limit. For $L = 2$, symmetry of the binomial coefficients with odd upper index gives $C_{n,2} = 2^{2n-1}$, yielding $2 \log 2 = 2H(1/2)$. ■

A.1 The Depth-one Endpoint

At depth one, W_1 is invertible almost surely, so the all-active mask is realizable: take $s = W_1^{-1} \mathbf{1}$. Its Jacobian is W_1 , whereas every other masked Jacobian satisfies $\|D_1 W_1\|_{2 \rightarrow 2} \leq \|W_1\|_{2 \rightarrow 2}$. Lemma 4 therefore gives

$$K_{n,1}(\alpha) = \|W_1\|_{2 \rightarrow 2} \quad \text{almost surely.}$$

The Gaussian spectral-edge limit (Bai and Yin, 1993) is $\|W_1\|_{2 \rightarrow 2} \rightarrow 2\sqrt{\alpha}$ almost surely. Thus $\mathbb{P}(K_{n,1}(\alpha) \leq 1) \rightarrow 1$ for $\alpha < 1/4$ and tends to zero for $\alpha > 1/4$, which proves $\alpha_1 = 1/4$. More generally, the layerwise bound $K_{n,L}(\alpha) \leq \prod_{\ell=1}^L \|W_\ell\|_{2 \rightarrow 2}$ shows that every $\alpha < 1/4$ belongs to the set defining α_L for each fixed L . Together with (40), this verifies that the set is a nonempty interval of the form $(0, \alpha_L)$ or $(0, \alpha_L]$.

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